

The empirical recurrence for OEIS A235655: recovering a conjecture that is not written down

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Abstract

OEIS A235655 records “Empirical recurrence of order 69”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 115$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235655 is “Number of $(n+1) \times (3+1)$ 0..1 arrays with the difference of the upper median and minimum value of each 2×2 subblock in lexicographically nondecreasing order rowwise and columnwise.”. It has offset 1 and begins

160, 1368, 11236, 85700, 627189, 4396769, 29920256, 198335901, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:42:15, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 200$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 115$ of the 200, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 115$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 400$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	20	c_{19}	505042808444	c_{37}	−60129429031186	c_{55}	−172971396808
c_2	−86	c_{20}	−548686466266	c_{38}	474894452979098	c_{56}	−198285283608
c_3	−808	c_{21}	−1956951457802	c_{39}	24173104994522	c_{57}	34502795298
c_4	7583	c_{22}	2713488219658	c_{40}	−407845840592251	c_{58}	30516877400
c_5	4300	c_{23}	6164719444740	c_{41}	3915924784132	c_{59}	−5395047780
c_6	−229906	c_{24}	−10302408695484	c_{42}	295652600366758	c_{60}	−3614927227
c_7	386502	c_{25}	−15905149358432	c_{43}	−16676008423884	c_{61}	646820962
c_8	3864099	c_{26}	31248107589017	c_{44}	−180639150007112	c_{62}	316739644
c_9	−12243876	c_{27}	33736500456754	c_{45}	16664564559922	c_{63}	−57351594
c_{10}	−40423326	c_{28}	−77357740059056	c_{46}	92770454814902	c_{64}	−19260969
c_{11}	198510196	c_{29}	−58842444712386	c_{47}	−11065661111696	c_{65}	3537654
c_{12}	258299908	c_{30}	158368357969796	c_{48}	−39883687965003	c_{66}	722920
c_{13}	−2154280400	c_{31}	84037451355478	c_{49}	5551851445432	c_{67}	−135184
c_{14}	−666557346	c_{32}	−270406563755797	c_{50}	14271631702581	c_{68}	−12560
c_{15}	17141501814	c_{33}	−97182468959038	c_{51}	−2190039141672	c_{69}	2400
c_{16}	−5217680646	c_{34}	387262778752426	c_{52}	−4217568563465		
c_{17}	−104727999808	c_{35}	88743151654576	c_{53}	688665589208		
c_{18}	77595998266	c_{36}	−466899735707345	c_{54}	1018661283398		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235655.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.